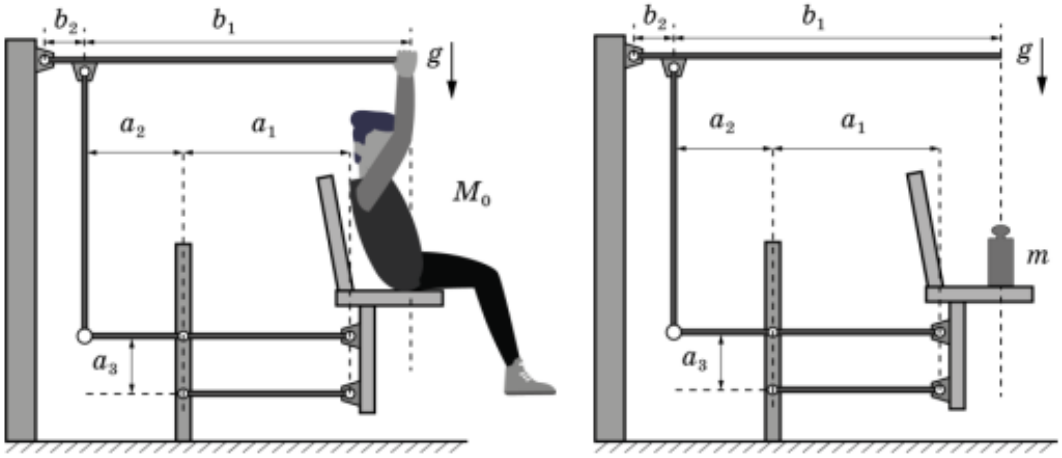
1- masala.

A-qism. Quvvat simulyatori. (5 ball)

Sport maydonchasida quvvat simulyatori rasmda ko‘rsatilgan. Kresloda o'tirgan sportchi richagga ***F*** kuch qo'llash orqali o'zini yuqori ko'taradi. Sistema richagi va sharniri tekis parallel harakatlanishini taminlaydi. Sportchi yo‘qligida simulyatorni muvozanatlash (yuqori richag gorizontal holatda bo‘ladi) m = 3,7 kg yukni qo'yishi kerak.



Massasi M_0 = 86 kg bo'lgan odam kresloda o'tirganda (oyog‘i erga tegmasdan) richagni gorizontal holatda ushlab turish uchunrichagga qanday vertikal F kuch bilan ta’sir qilishi kerak?

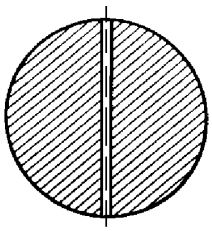
Hisoblashda tutqichlarning uzunliklari: a_1 = 27,5 sm; a_2 = 13,0 sm; a_3 = 17,5 sm; b_1 = 73,5 sm; b_2 = 8,5 sm. Erkin tushish tezlashishi g =10 m/s^2.

B-qism. Bor modeli. (5 ball)

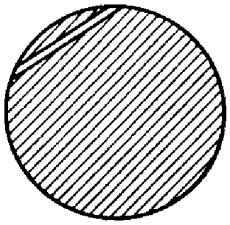
Vodorod atomining n-orbitasidagi elektronning kinetik va potensial energiyasini toping. Masalani n=1, 2, 3 hollari uchun yeching. h=6,626·10^{-34} J·s, m=9,11·10^{-31}kg, r_1=0,529·10^{-10} m.

2- masala. Kichik sayyoradagi shaxta (8-ball)

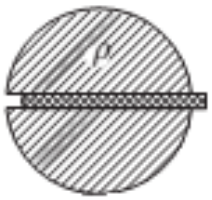
a) Biror sayyoraning diametri bo’ylab ingichka shaxta qazildi. Bunda shaxtaning bir uchidan qo’yib yuborilgan kichik tosh qanday davr bilan tebranadi? $\boldsymbol{\rho}$ – sayyoraning zichligi.



b) Huddi shu sayyorada chizmada ko’rsatilgandek ixtiyoriy vatar bo’ylab, to’g’ri shaxta qazildi va shaxtaning bir uchidan tosh qo’yib yuborildi. Bu toshning tebranish davrini toping. $\boldsymbol{\rho}$ – sayyoraning zichligi.

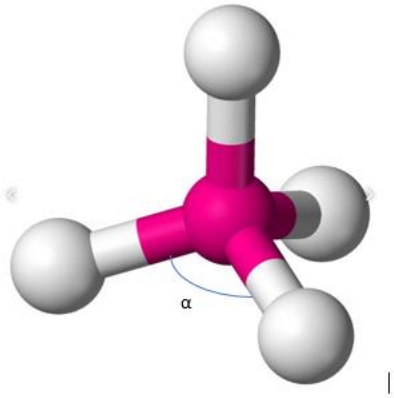


c) Sayyoraning diametri bo’ylab qazilgan shaxtaning ichiga zichligi sayyoraning zichligi bilan bir hil bo’lgan materialdan tayyorlangan sterjen joylashtirildi. Bu sterjenning kichik tebranishlar davrini toping. $\boldsymbol{\rho}$ – sayyoraning zichligi.



3-masala. Metan molekulasi (8-ball)

Metan molekulasi CH_4 to‘rtta vodorod va bitta uglerod atomidan tashkil topgan. Quyidagi rasmda bitta metan molekulasining fazoviy strukturasi ko‘rsatilgan:



Qizil rangda C uglerod atomi (tashqi qatlamida to’rtta elektron bor), oq rangda esa H (bir elektronli atom) vodorod atomi tasvirlangan. Metan molekulasidagi ikkita qo‘shni C-H bog‘lar orasidagi burchaklarini toping.

4-masala. Fotonli “gaz” (14 ball)

Kvant fizikasi nuqtai nazaridan elektromagnit nurlanish xaotik harakatlanayotgan va o‘zaro tasirlashmaydigan zarrachalar-fotonlar toplamidan iborat.

Boshqacha aytganda elektromagnit nurlanish bu molekulyar–kinetik nazariyada ko‘riladigan ideal gazga ko‘p jihatlari bilan o‘hshash fotonlardan tashkil topgan gazdir. Muhim farqlari ham bor. Hamma fotonlar bir hil tezlikda (yorug‘likning vakuumdagi tezligi bilan) harakatlanishadi, va ularning soni doimiy saqlanmaydi, holat ozgarganda fotonlar tug‘ilishi va yutilishi mumkin. Shunga qaramay foton gazining qator hossalarini ideal gazning molekyular - kinetik nazariyasiga asoslanib aniqlash mumkin va bu masalada shuni bajarish taklif qilinadi.

1. Ideal gaz zarrachalarining yassi sirtga bosimi P quyidagi formula bilan aniqlanishini isbotlang:

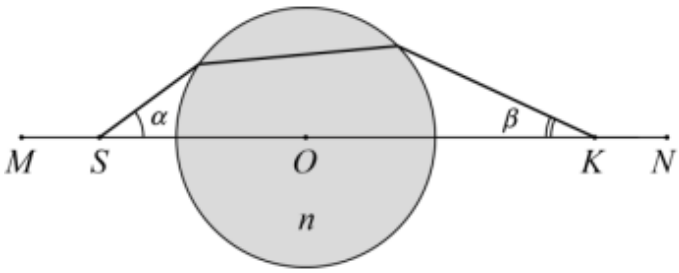
$P=\frac{1}{3}n<pv>$ (1),

Bu yerda n – hajm birligidagi zarrachalar soni, $\vec{\mathbf{v}}$ - zarrachalar tezligi, $\vec{\mathbf{p}}$ - ularning impulsi, <pv>- skalyar kopaytma ($\vec{\mathbf{v}} \cdot \vec{\mathbf{p}}$) ning o‘rtacha qiymati.

2. Ideal gaz bosimining formulasidan foydalanib yorug‘likning bosimini
   $P=\frac{1}{3}u$, (2)
   formula yordamida hisoblash mumkin ekanligini isbotlang, bu yerda u- nurlanish energiyasining hajmiy zichligi.
3. Temperatura va hajmning kichik o‘zgaradigan holidagi foton gazi uchun Karno siklini o‘rganib yorug‘lik bosimini absolyut haroratning to‘rtinchi darajasiga proporsional ekanligini isbotlang.
4. Avvalgi punktdagi natijalardan foydalanib absolut qora jismning birlik yuzasidan nurlanishi quvvati uchun Stefan-Boltsman qonunini: $W=\sigma T^4$ keltirib chiqaring, bu yerda σ-Stefan-Boltsman doimiysi, T- absolyut harorat. Bosim va haroratning to‘rtinchi darajasi orasidagi proporsionallik koefitsienti bilan Stefan –Boltsman doimiysi σ orasidagi bog‘lanishni toping. Bog‘lanishni keltirib chiqarishda devorning birlik yuzasi bilan vaqt birligida to‘qnashayotgan gaz zarrachalarining soni $\nu=\frac{1}{4}n<v>$ ga teng ekanligini inobatga oling, bu yerda n-hajm birligidagi zarrachalar soni, <v>-zarrachalar tezlik modulining o‘rtacha qiymati.
5. Foton gaz ustida bajarilayotgan siklning foydali ish koeffisientini ifodasini toping. Sikl to‘rtta ketma-ket jarayondan tashkil topgan:
   1) T_1 temperaturali holatdan izobarik kengayish,
   2) Temperaturasi T_2 holatga $PV^{4/3}$=const qonun bo‘yicha o‘tish,
   3) Izobarik siqilish,
   4) Boshlangich holatga yana $PV^{4/3}$=const qonun boyicha o‘tish.

5-masala. Shaffof shar (10 ball)

S nuqtaviy yorug‘lik manbai tomonidan chiqarilgan yorug'lik nurlari sindirish ko‘rsatkich *n* bo‘lgan bir jinsli shaffof sharga tushmoqda. Nur shar markazidan o‘tuvchi *MN* to‘g‘ri chiziqqa nisbatan α burchak ostida tushib ikkita sinishdan keyin sharning chegarasini kesib *K* nuqtada *MN* to‘g‘ri chiziqqa nisbatan β burchkak ostida burchak tushadi. (rasmga qarang.). Masofa *SK* = *l*.



1) Shar radiusi *R* ni va *SO* masofani *l*, *α*, *β* , *n* parametrlar orqali ifodalang. Manbadan sharning markazigacha bo'lgan masofasini ifodalangl,

2) *R* va *SO* masofalarni *n* = 2, *α* = 60, *β* = 30, *l* = 10 sm qiymatlar uchun hisoblang.